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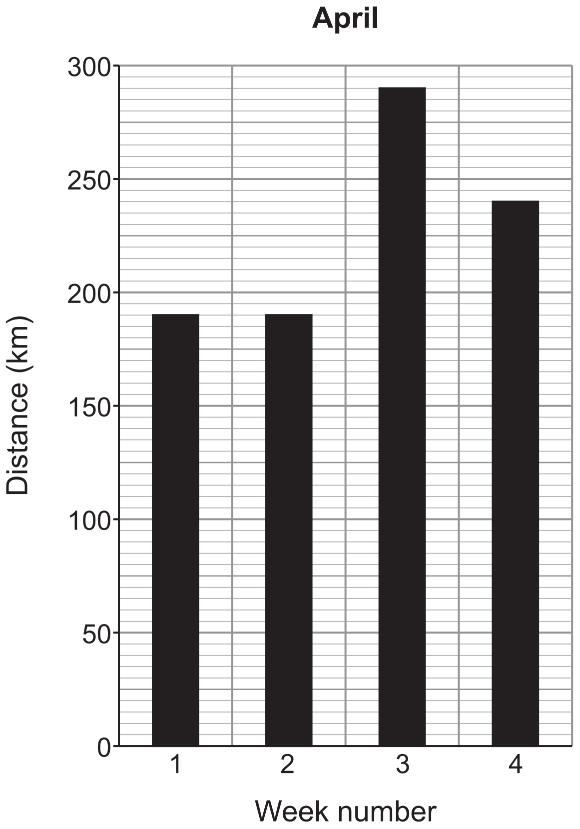
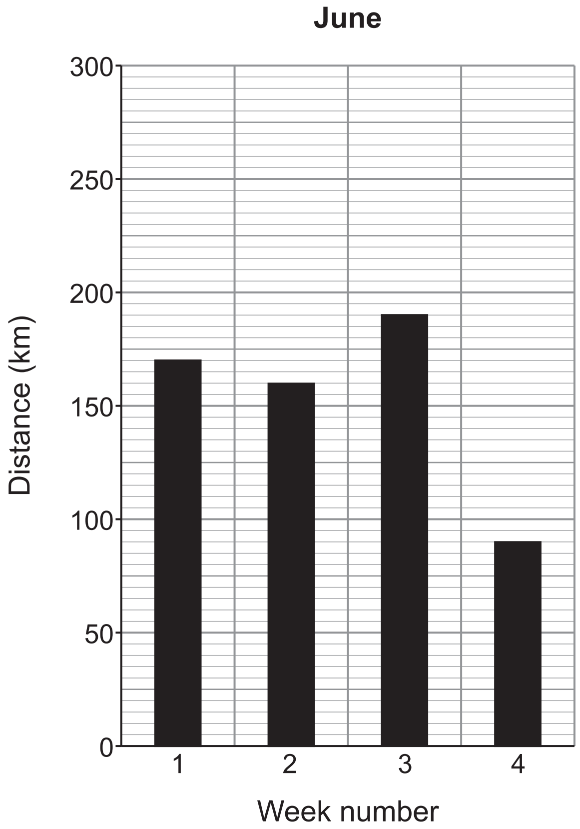
4. A fitness app is used by a cyclist to monitor his cycling performance.

(a) The distance cycled each week is displayed as a chart.
The data for two months, April and June, are shown below.





(i) The total distance travelled by the cyclist in June was 610 km.
Show that the total distance travelled by the cyclist in April was 910 km. [1]

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(ii) The cyclist cycled for a total time of 31.5 hours during April and 19.25 hours during June.
A student suggests that the mean speed of the cyclist is greater in June than in April.
Use an equation from page 2 to determine whether the student is correct. [3]
Space for calculations.

....

....

(b) (i) A track cyclist has an initial velocity of 8.0 m/s.
She uniformly accelerates towards the finish line, travelling a distance of 42 m, in a time of 3.0 s. [4]

Use the equation:

$$x = ut + \frac{1}{2}at^2$$

to calculate the cyclist's acceleration and state its unit.

acceleration = ...

unit = ...

(ii) The resultant force that causes the acceleration towards the finish line is 284 N.

I. Use an equation from page 2 to calculate the mass of the bike and the cyclist. [2]

mass of bike and cyclist = ... kg

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II. The mass of the cyclist is 64.5 kg. Calculate the mass of the bike. [1]

mass of bike = ... kg

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